

Omar Baruch M. Lopez

AI Engineer II | Applied AI & Data Engineering | Industrial AI | Enterprise AI Architecture Focus | MBA Candidate

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PROFESSIONAL SUMMARY

AI Engineer II at Bosch Mexico with 4+ years building AI-driven solutions, data engineering workflows, analytics capabilities and workflow automation for production, operations, finance/controlling and business environments. Hands-on work includes SAP-to-modern-data integration, ETL pipelines, Power BI reporting, ML MVPs, APIs, runbooks and reusable documentation. Current GenAI work and learning focus includes prompt engineering, AI assistants, text/image/video generation, AI agents and evaluation. Robotics engineering, Dual MBA studies and active TOGAF Enterprise Architecture certification preparation support work at the intersection of industrial AI, enterprise architecture, governance and business-aligned solution design.

CORE SKILLS

AI, ML & Automation

AI engineering, Machine learning, Python, TensorFlow, Scikit-learn, Computer vision, ML MVPs, Automation

Data Engineering & Analytics

SQL, MySQL, Oracle SQL, SAP data integration, SQL Server, ETL, Alteryx, Power BI, DAX, Data Lake workflows, Data pipelines, Reporting

GenAI / AI Tooling Focus

Generative AI, Prompt engineering, ChatGPT, OpenAI Codex, Anthropic Claude, AI agents, RAG, Text generation, Image generation, Video generation, Multimodal AI, Evaluation, Responsible AI

Software, Integration & Platforms

APIs, Flask, FastAPI, Airflow, AWS EC2, Elastic Beanstalk, Azure, SageMaker fundamentals, Postman, GitHub, GitLab, Jira, PyTorch, Hugging Face, Documentation

Architecture & Business

Solution architecture, Enterprise architecture, TOGAF certification preparation, Business capability alignment, Governance, Roadmapping, Stakeholder management, Technical leadership, Business strategy, Finance basics

Languages

- Spanish: Native
- English: Full professional proficiency, supported by TOEFL iBT 78 and bilingual professional work.

PROFESSIONAL EXPERIENCE

Bosch Mexico

Guadalajara, Mexico
2022 - Present

AI Engineer II, Jan 2026 - Present

AI Engineer, 2022 - Feb 2026

- Delivered AI-enabled and data-driven solutions for production, operations, finance/controlling and business environments, translating stakeholder needs into technical assets.
- Built production analytics and data-pipeline workflows across internal Bosch initiatives including TLP Scrap V2, OPC/TNS, AUC Tracking, Operational Cockpits and Productivity Bonus, using Alteryx, SQL Server, Power BI and SAP/data-lake sources.
- Implemented integration and workflow-enablement assets including the MIURA FastAPI service, Fraccion Arancelaria customs/tariff-classification data enablement, and Airflow/SFTP/REST API automation.
- Created ML/AI MVPs and PoCs across vehicle detection, staff mobility optimization, production-target regression, object detection and purchase-order anomaly detection using Python, scikit-learn, NetworkX, PyTorch/TensorFlow, YOLOv5/Detectron2 and APIs.
- Authored runbooks, documentation, validation logic, manuals and training material to transfer knowledge, improve stakeholder adoption and structure ambiguous requirements into reusable workflows.

Cyrus Global Capital

United States / Remote
Jul 2024 - Present

Co-Founder & Chief Financial Officer

- Contributes to institutional structure, financial coordination, internal reporting discipline and long-term operating perspective for a private capital initiative.
- Applies data, governance, business strategy and finance fundamentals to reporting and operational structure within a private capital initiative.

Bosch Mexico

Mexico

Apr 2019 - Dec 2021

Mexcellence Scholarship Program

- Completed a formative Bosch-related scholarship stage that bridged robotics studies with professional exposure and applied AI/IoT project work.

EDUCATION

Dual MBA - Master of Business Administration / Maestria en Direccion y Administracion de Empresas

MIU City University Miami and UNIR Mexico | Jul 2025 - Mar 2027 | In progress

Licenciatura en Ingenieria Robotica

Universidad de Guadalajara | Jan 2017 - Jun 2021

CERTIFICATIONS & TRAINING

- Introduction to Generative AI - Duke University / Coursera, May 2025
- AI Agents for Cognigy Users - NiCE Cognigy, Sep 2025
- Cognigy Foundations (2025) - NiCE Cognigy, Sep 2025
- Machine Learning Data Lifecycle in Production - DeepLearning.AI, May 2023
- Introduction to Machine Learning in Production - DeepLearning.AI, Mar 2023
- Deploy Models with TensorFlow Serving and Flask - Coursera, Jan 2023
- AWS Elastic Beanstalk: Deploy a Python (Flask) Web Application - Coursera, Jan 2023
- AWS Shared Responsibility Model, Amazon Athena, Kinesis Streams and SageMaker fundamentals - AWS, 2024
- MySQL 8.0 - Sulens / BYW Consulting Services, 2022
- Agile with Atlassian Jira - Atlassian / Coursera, Mar 2022
- Principles of Leadership: Leading Technical Teams Specialization - University of Colorado Boulder, Mar 2025
- Introduction to Finance: The Basics - University of Illinois Urbana-Champaign, Mar 2025
- Lean Six Sigma White Belt - International Lean Six Sigma Group, Nov 2025
- Leading People - Harvard Business Impact, May 2026
- TOGAF Enterprise Architecture certification preparation - In progress
- Bosch internal training portfolio - information security, data protection, compliance, CIP, Power BI, applied ML, ABAP basics, supply-chain security and process standardization, 2022 - 2025
- Professional engagements - Talent Land Guadalajara Explainable AI, Connectory Talks Data Analytics, Mentoring4GS and robotics workshops, 2019 - 2025
- TOEFL iBT Score 78 - ETS, Sep 2020